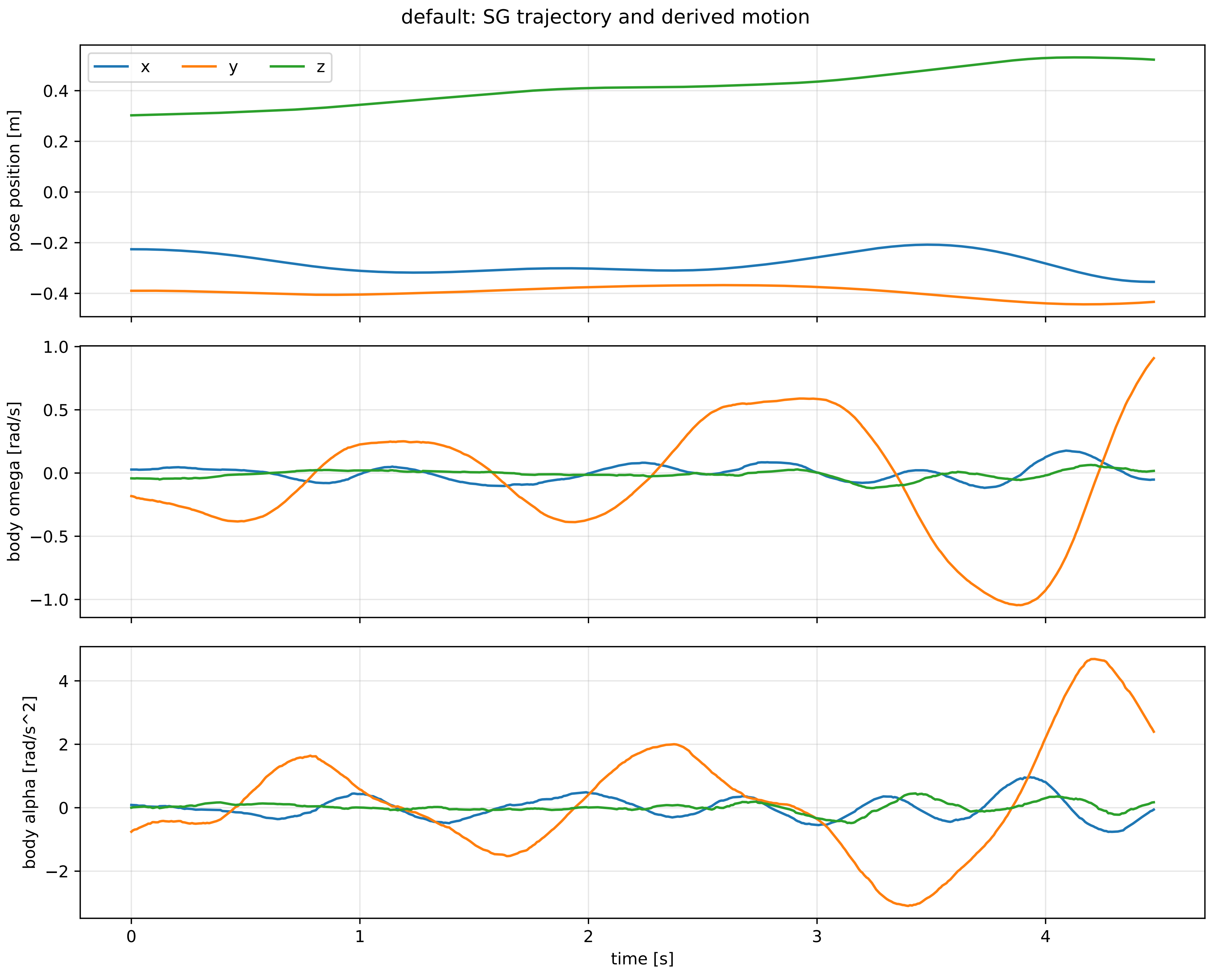
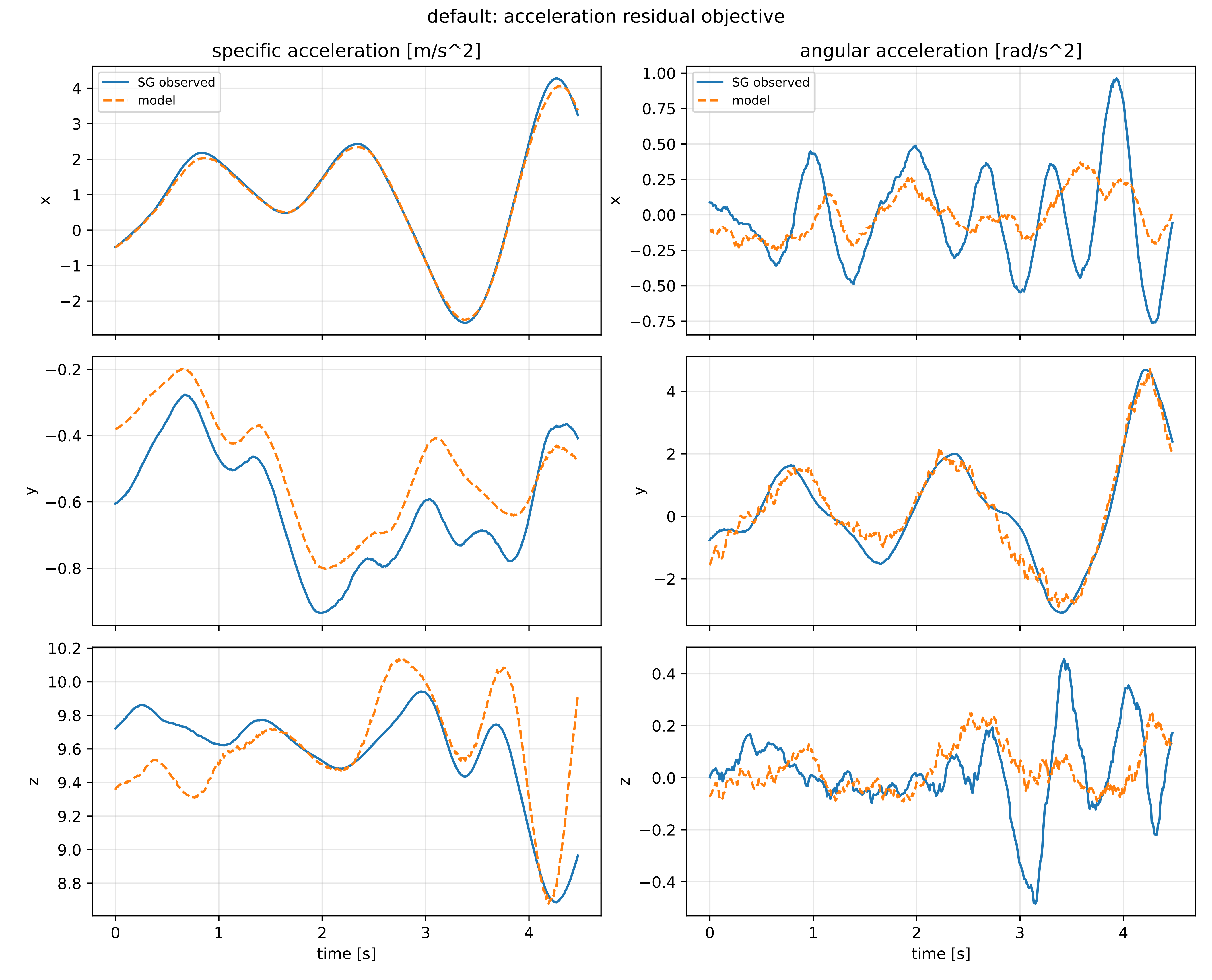
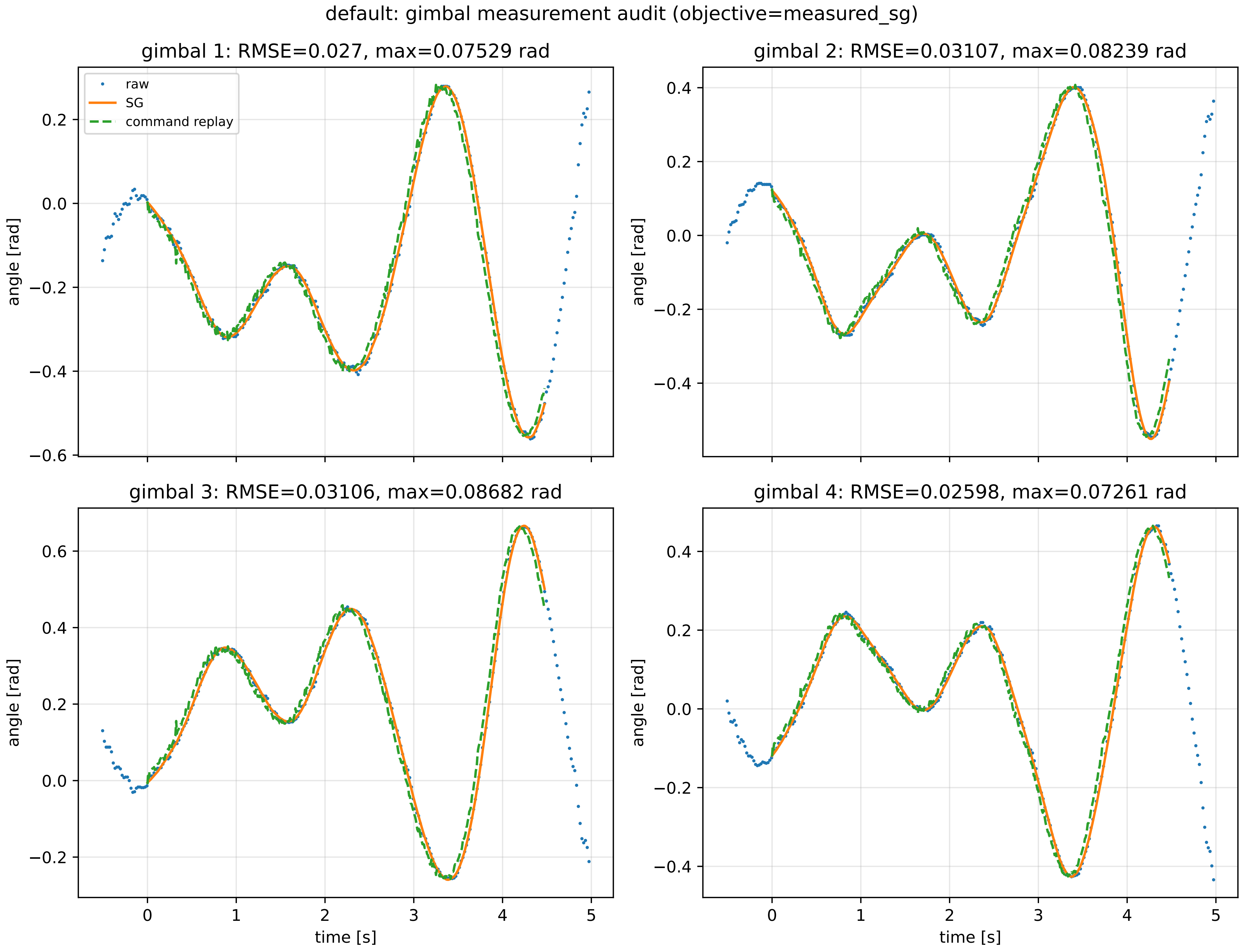
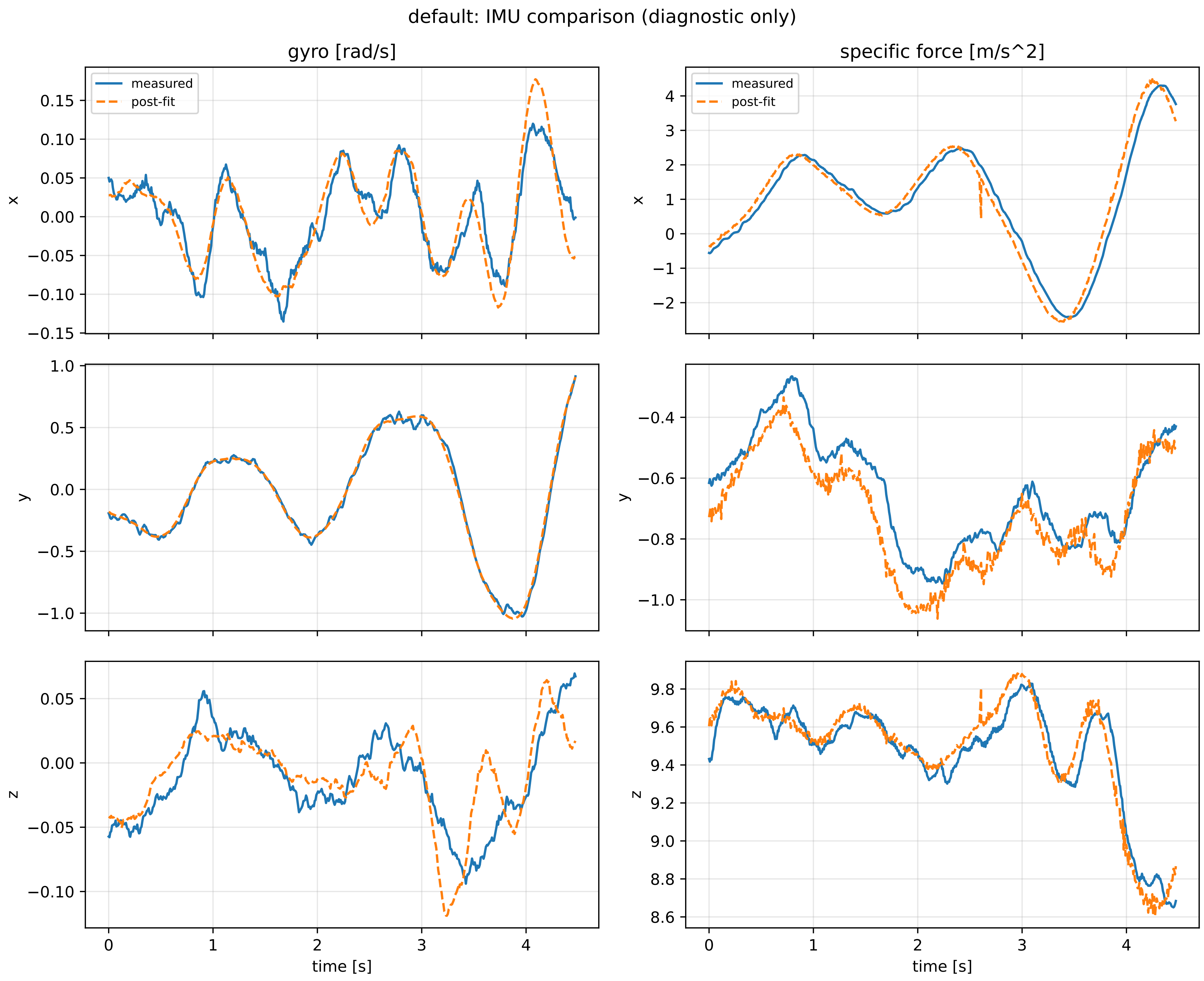
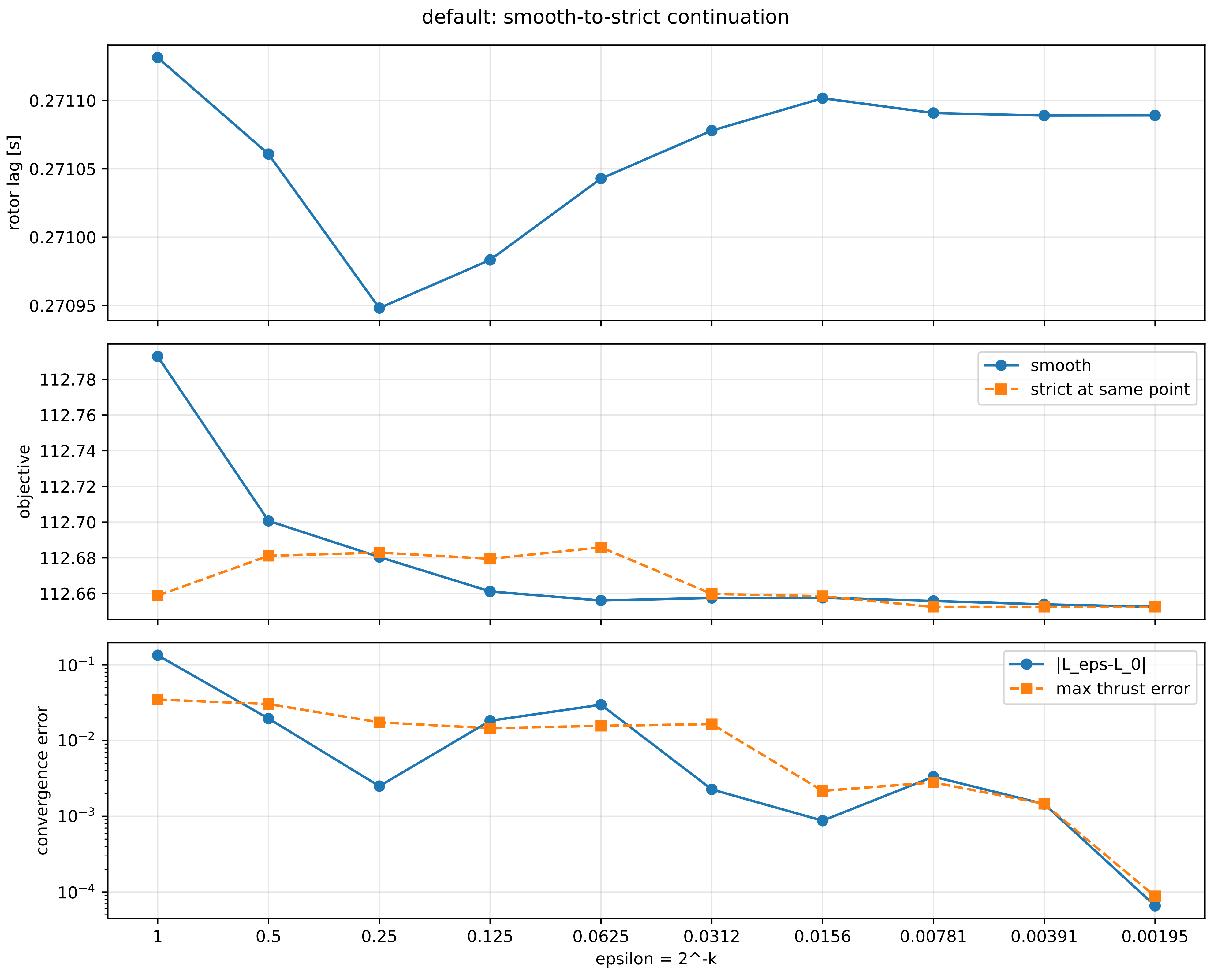

default: SG trajectory and derived motion
x
y
z
pose position [m]
0.4
0.2
0.0
−0.2
−0.4
body omega [rad/s]
1.0
0.5
0.0
−0.5
−1.0
body alpha [rad/s^2]
4
2
0
−2
0
1
2
3
4
time [s]


default: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
SG observed
model
x
y
z
x
y
z
time [s]
time [s]



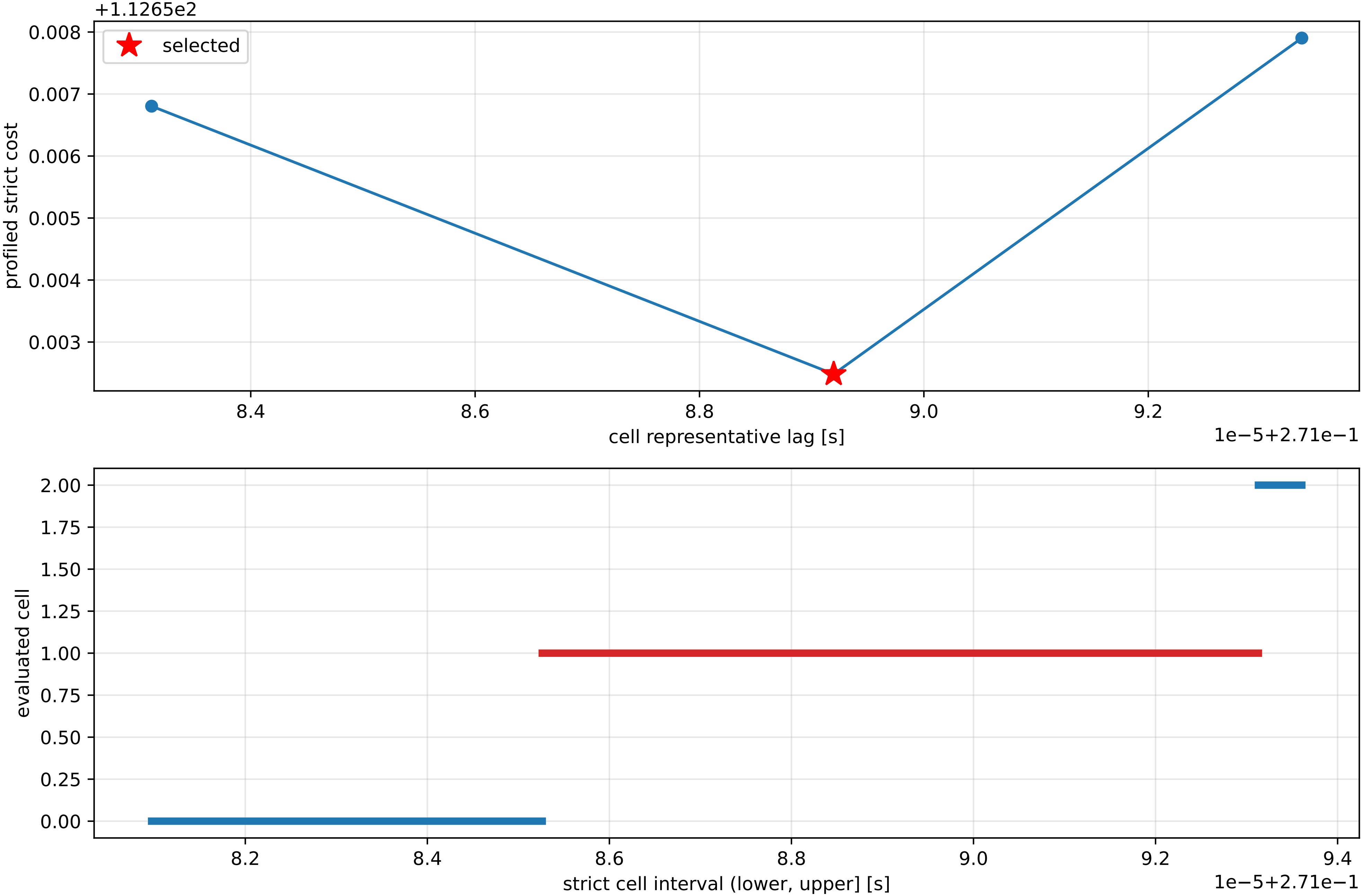
default: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]


default: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



default: smooth-to-strict continuation
rotor lag [s]
0.27110
0.27105
0.27100
0.27095
objective
112.78
112.76
112.74
112.72
112.70
112.68
112.66
smooth
strict at same point
convergence error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

default: exact strict-ZOH lag cells

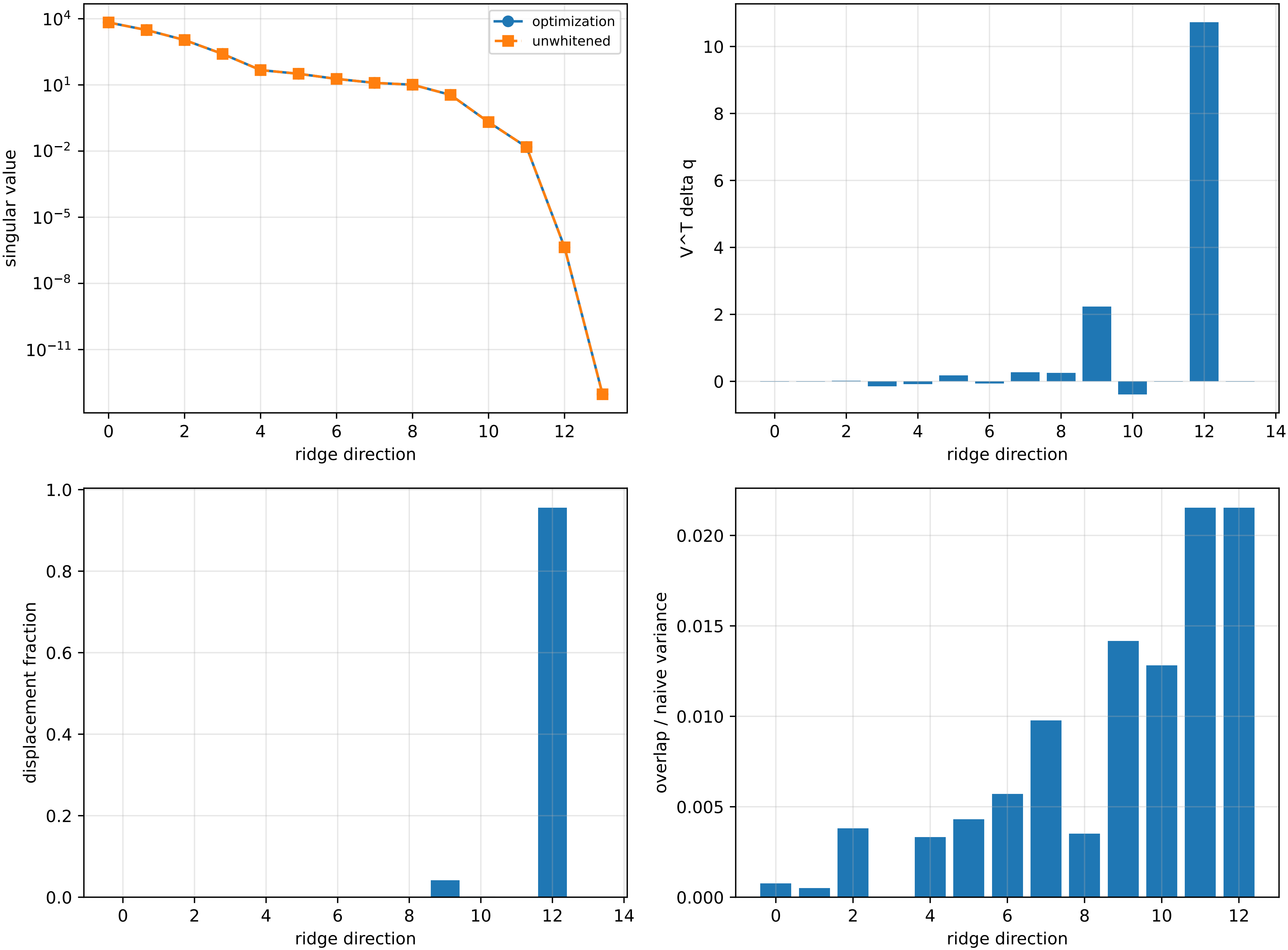


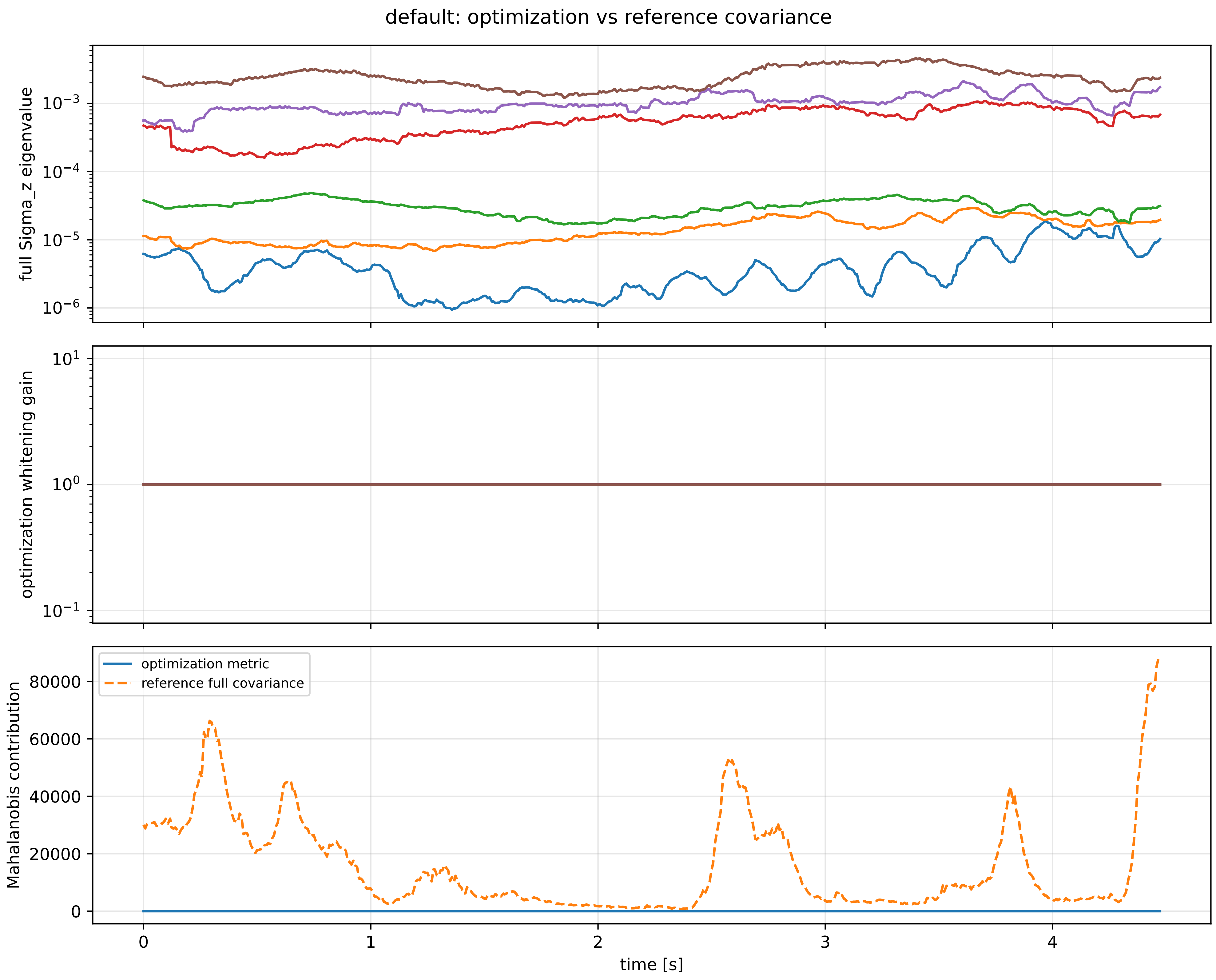
selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

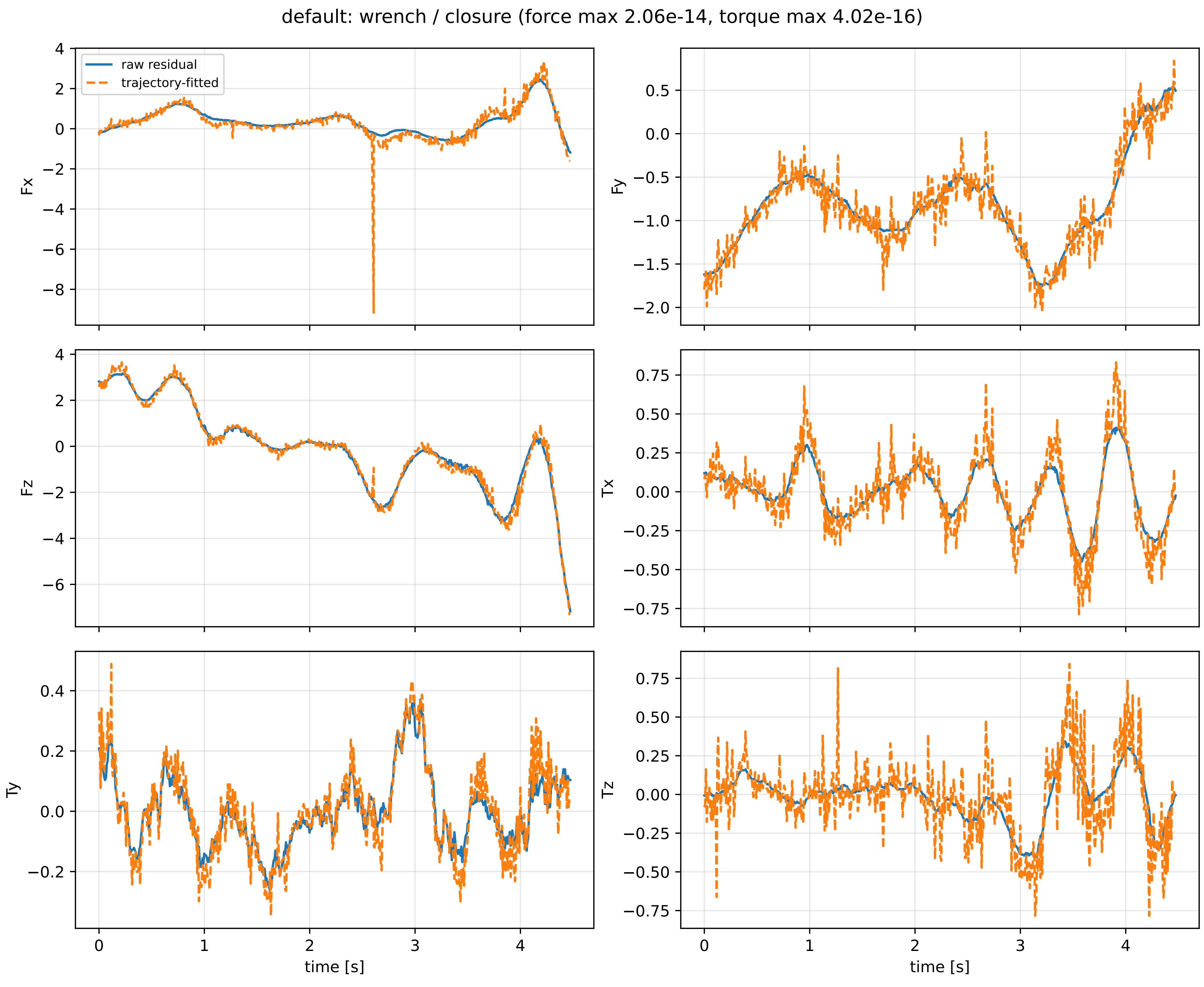
default: parameter estimate

```
Case: default
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.52043204
inertia [kg m^2]:
[[ 0.55443709  0.01372501 -0.00323582]
 [ 0.01372501  0.26168335 -0.08172245]
 [-0.00323582 -0.08172245  0.79125006]]
CoG body [m]: [-0.026435    0.01133929  0.00941535]
force effectiveness: [2.87131373 2.32387855 2.56341484 3.07927654]

scale-free J/m [m^2]:
[[ 0.0737241   0.00182503 -0.00043027]
 [ 0.00182503  0.03479632 -0.01086672]
 [-0.00043027 -0.01086672  0.10521338]]
scale-free f/m: [0.3818017  0.30900865 0.34086005 0.40945474]

rotor lag cell: (0.271085262, 0.27109313] s
representative: 0.271089196 s
```

default: ridge spectrum (rank 13, nullity 1, ||Jv||=1.61e-13)




default: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


default: wrench / closure (force max 2.06e-14, torque max 4.02e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]
0
1
2
3
4
4
2
0
−2
−4
−6
−8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.4
0.2
0.0
−0.2